



VIDEM



In a busy urban clinic, healthcare professionals often face the challenge of **rapid diagnosis** .

With VIDEM, clinics can provide immediate, precise care without the need for costly, complex lab setups, enhancing both patient outcomes and operational efficiency.

About Our Team

01



Abhay Bhat

Master of Engineering in
Bioengineering 2025



Anh Doan

Master of Translational
Medicine 2025



Carlos Mauricio Neyra

Masters of Law (LLM)



Gracie Teed

Master of Engineering in
Bioengineering 2025



Anderson Wang

Master's of Science in
Materials Science and
Engineering

Problem

02

“Time of diagnosis to time of care can always be faster for patient diagnosis. Especially with delays in handoff”

-Caesar Djavaherian, CIO of Carbon Health



Patient visit and patient care

Critical Public Health Crisis

The United States is facing a persistent rise in bacterial and viral infections, compounded by barriers in accessibility, diagnosis, and delayed treatment.

Limited Access to Testing

Key challenges include limited access to timely and accurately testing. Current point of care diagnoses are high cost.

Delayed Results and Missed Treatments

This can lead to missed treatment opportunities. Many infections, especially STIs can be asymptomatic.

Growing Resistance

The crisis is further exacerbated by growing antimicrobial resistance and increased emergency situations involving infections such as sepsis and meningitis.



“ Delaying treatment can increase the likelihood of STI transmission by up to 10 times and prolong infections ” (Guddu et al., 2024).

Solution

03



Time Saving Point-of-Care Diagnostic Device

1. Innovative Sensor

This tech is a biosensor that uses acoustic waves to detect the presence of bacterial or viral DNA + Proteins. Acoustic wave sensors have a low limit of detection and low cost.

2. Rolling Circle Amplification (Isothermal)

Paired with RCA, this is a point-of-care solution providing results within 30-35 minutes.

3. Multiplex Testing

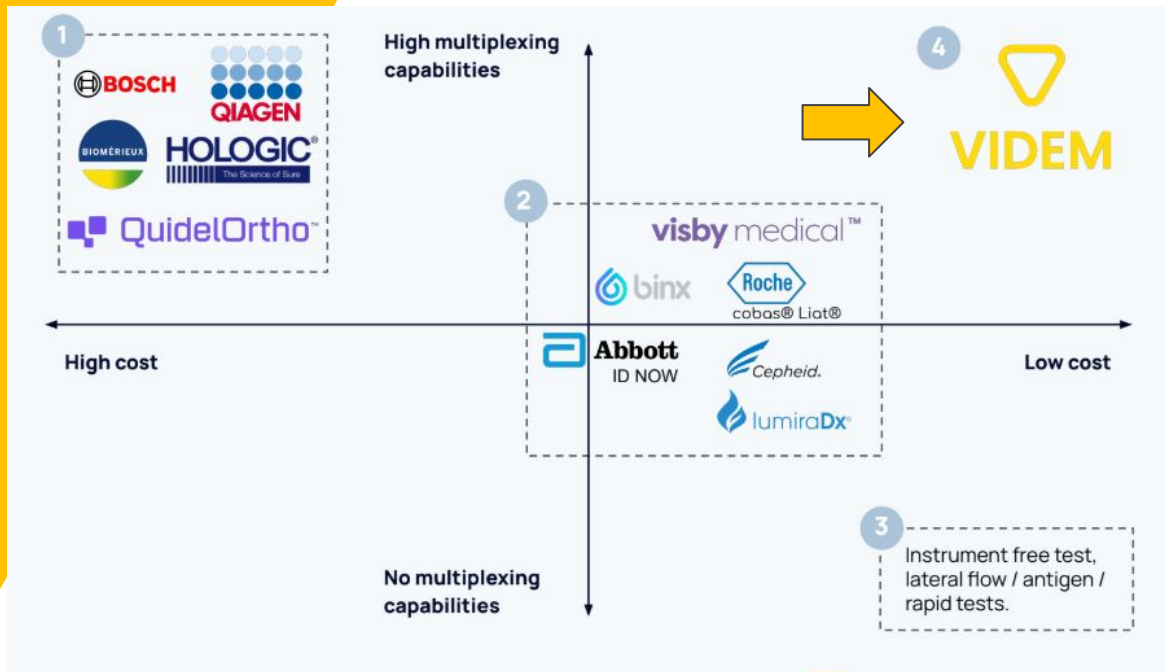
The device can provide low-cost and highly accurate diagnosis of up to 10 diseases per test, suitable for time sensitive testing and faster time-to-patient care.

Competition

04



Videm's Competitive Analysis



Efficiency:

- Speed and sensitivity
- Cost efficiency
- Short turnaround time for results

Versatility

- Multiplexed syndromic testing
- Portable and point-of-care use

User Friendly

- Simple and user-friendly design
- Lower sample volume requirement

Syndromic testing and multiplexing differentiate Videm in Emerging Competitive Market

	VIDEM	Avianna	Visby
Functionality	Syndromic testing, reusable cartridges.	Single-disease detection for emergencies.	Single-use PCR tests for rapid diagnostics.
Multiplexing	Multi-pathogen detection.	Limited to single-target tests.	Limited multiplexing (e.g., COVID-19, flu).
Technology	RCA + SAR sensors for high sensitivity.	Biochip; portable but less precise.	Miniaturized PCR; no instruments needed.
Target Market	Clinics, urgent care, retail health.	Military and austere environments.	Clinics, resource-limited settings.
Applications	STIs, respiratory, GI diseases.	Lyme disease, COVID-19, trauma.	STIs, respiratory diseases.
Turnaround Time	<30 minutes.	10–20 minutes.	~30 minutes.
Portability	Portable, clinic-ready device	Ultra-portable; mobile integration	Palm-sized, instrument-free.

IP Control and Analysis

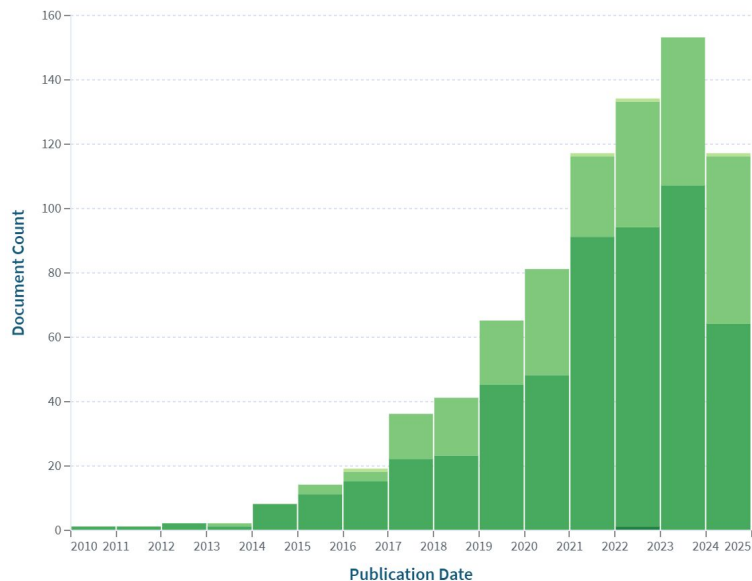
05



Similar Technical Solutions - IP Search

Search query: Graphene AND (biosensor AND diagnostic AND ("Point of care" AND disease)))

Number of Patents



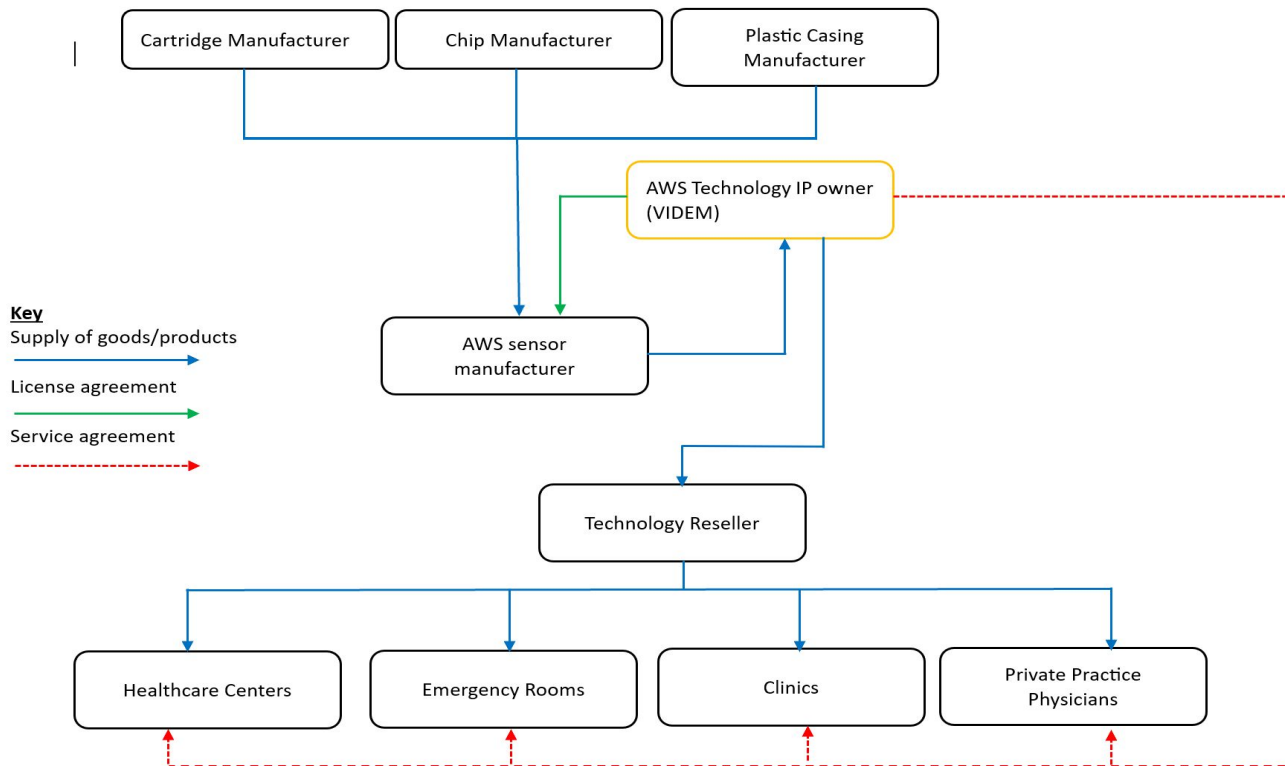
Document Type

Amended Application
Granted Patent
Patent Application
Search Report

Top Applicants

 Mazed Mohamm... 51	 Univ California 29	 Miroculus Inc 19	 Quantum Si Inc 18	 Univ Northwestern 15
 Juno Diagnostics... 14	 Univ Illinois 13	 Univ Northeastern 11	 Univ Duke 11	 Postrel Richard 11
 Dexcom Inc 11	 Univ Pittsburgh ... 10	 Univ Michigan Re... 10	 Univ King Abdull... 10	 Dullen Deborah 9

Licensing - Value Chain



Best Application Position for Videm

06



There is a need for fast and accurate test results in many sectors of healthcare



Current Target

Videm is currently targeting STI screening.



Market potential for STI screening

TAM ~\$12 billion

SAM - US accounts for 40-50% of the global diagnostics market

SOM - \$600 million if Videm can capture 10% of the market

Emergency Bacterial Infection Diagnostics

"In emergencies like meningitis or sepsis, doctors need results they can act on right away—ideally something they can use at the bedside. For other conditions though, they might care more about getting detailed, accurate information over how fast it comes in."

- Carbon Health Chief Innovation Officer and Emergency Physician

What does this include?

Meningitis, MRSA, Sepsis, etc.

Easier market entry

Less saturated market

Different regulations

Market Potential - Emergency Bacterial Infection

TAM

over \$16 billion

SAM

\$4.5 billion

SO

\$32-425 million

Market Applications

07



Healthcare Market Applications

"The bottleneck is access, not urgent time of care"
-Tang Center
Nurse Practitioner

STI

Largest TAM of bacterial infections, but relatively **mature space** with **satisfactory diagnosis** tools, and the cost benefit of the device is negated by the current industry.

Sepsis

Would be extremely **useful** to target this class of **infections for emergency room**, but biomarkers are unclear and the **device would be much more complex and expensive**

Meningitis

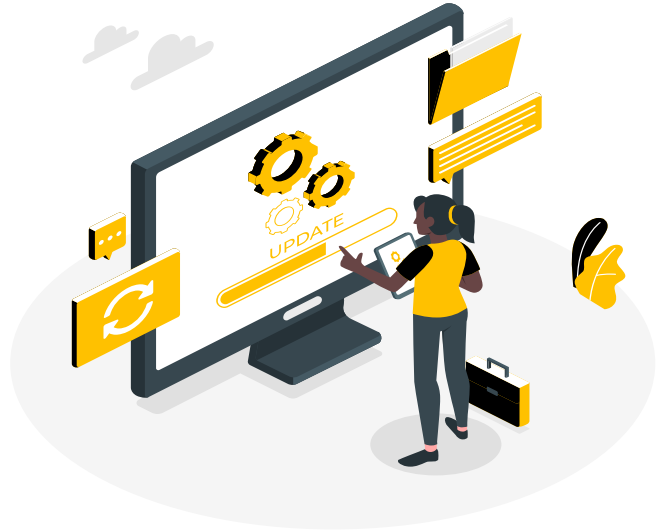
Medium sized market in the U.S. with a need for faster diagnosis for life threatening cases. **Niche and easy point of entry, but completely different set of regulations**

Respiratory Tract Infections

Large market, but other methods without Videm's tech can address

Regulatory Considerations (US)

08



Regulations for new diagnostic devices are determined by:

1. Device intended use
2. Risk classification
3. Technological characteristics

Class II

- Diagnostic tests for STDs like chlamydia and gonorrhea
- Devices designed to detect and identify microbial pathogen nucleic acids in cerebrospinal fluid (CSF)

Class III

- Tests for HIV due to the critical nature of accurate diagnosis.



Regulatory Strategies for Videm

If Videm pivots into **bacterial infection diagnostics**, we recommend:

- Determining a sufficient predicate device for **510(k)** filing such as:
 - **QIAGEN Meningitis/Encephalitis Panel**
 - FDA Approved November 4, 2024
 - real-time PCR simultaneous detection of multiple viral, bacterial, and fungal pathogens for **meningitis** and **encephalitis**
 - delivers results in about **one hour**

If Videm stays with **STI diagnostics**, we recommend:

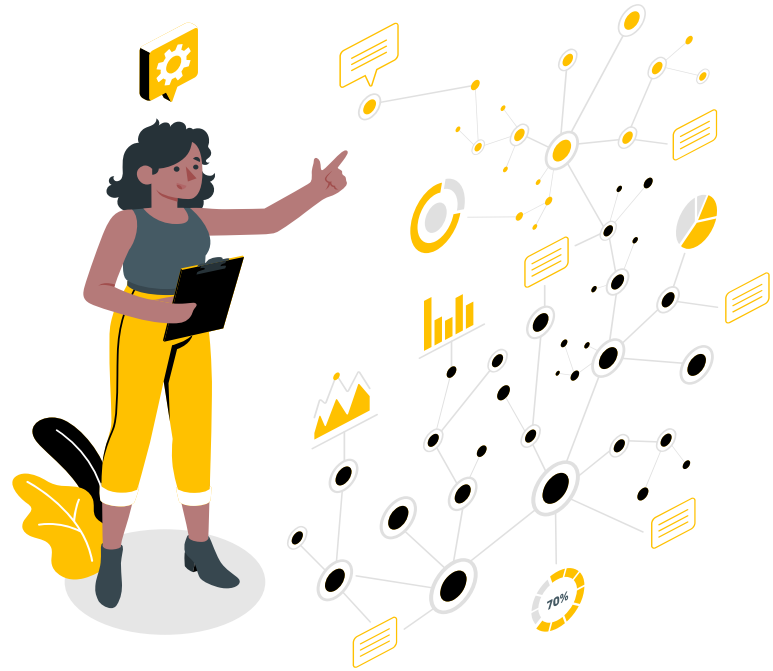
- Determining a sufficient predicate device for **510(k)** filing such as:
 - **Visby Medical Sexual Health Test**
 - FDA Approved March 14, 2023
 - a PCR diagnostic test for the detection of STIs (**Chlamydia**, **Gonorrhea**, and **Trichomoniasis**)
 - Deliver results **under 30 minutes**

“Begin Q-Sub Process ASAP!” - Peter Coronado

Pre-Submission > Submission Issues Request (SIR) > Study Risk Determination > Informational Meeting

Overall Evaluation

09



Videm should PIVOT their focus within the healthcare field towards emergency room bacterial infections

PROS	CONS	Market Entry
Faster Easier accessibility Low cost Multiplexing	Saturated STI market that is barricaded by FDA and insurance Similar solutions exist that utilize similar technology	Easier entry in a smaller market that is less dominated <u>and</u> with better use cases where fast results are more essential

Thank You



Appendix

Regulation requirements for new diagnostic devices are determined by:

1. Device intended use
2. risk classification
3. technological characteristics

Diagnostic devices designed to detect and identify microbial pathogen nucleic acids in cerebrospinal fluid (CSF) are classified as **Class II*** medical devices. More specifically, the FilmArray® Meningitis/Encephalitis (ME) Panel, which identifies multiple pathogens from CSF samples, was classified as a Class II device through the De Novo classification process.

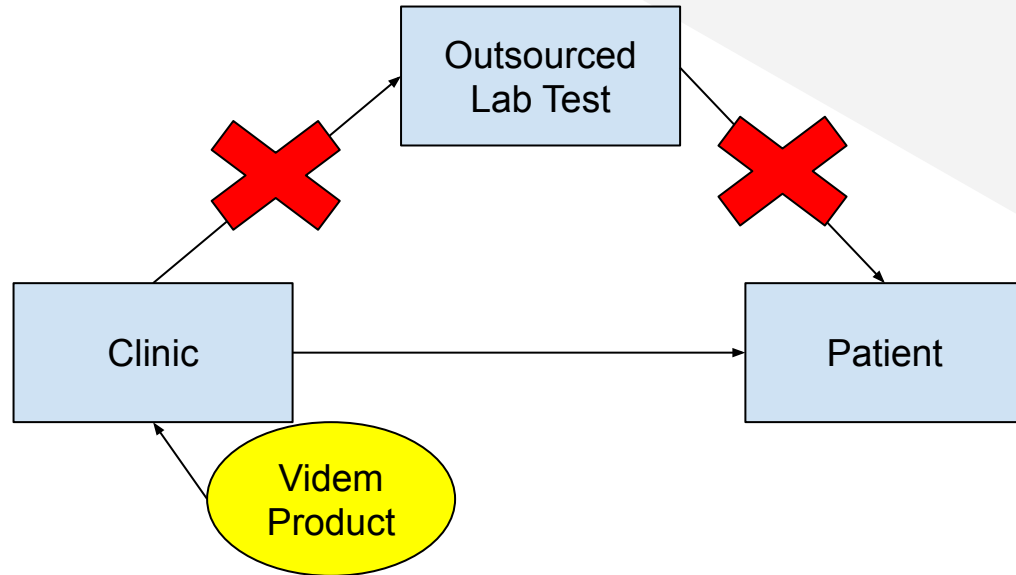
For reference, diagnostic tests for STDs like **chlamydia and gonorrhea are commonly classified as Class II** devices

Diagnostic tests for **HIV are often classified as Class III**** devices due to the critical nature of accurate diagnosis.

***Class II:** These devices are considered moderate-risk and, in addition to general controls, are subject to special controls, which may include specific labeling requirements, mandatory performance standards, and postmarket surveillance.

****Class III:** Class III devices are deemed high-risk and typically support or sustain human life, are of substantial importance in preventing impairment of human health, or present a potential unreasonable risk of illness or injury. These devices require premarket approval (PMA), demonstrating safety and effectiveness through rigorous scientific evidence.

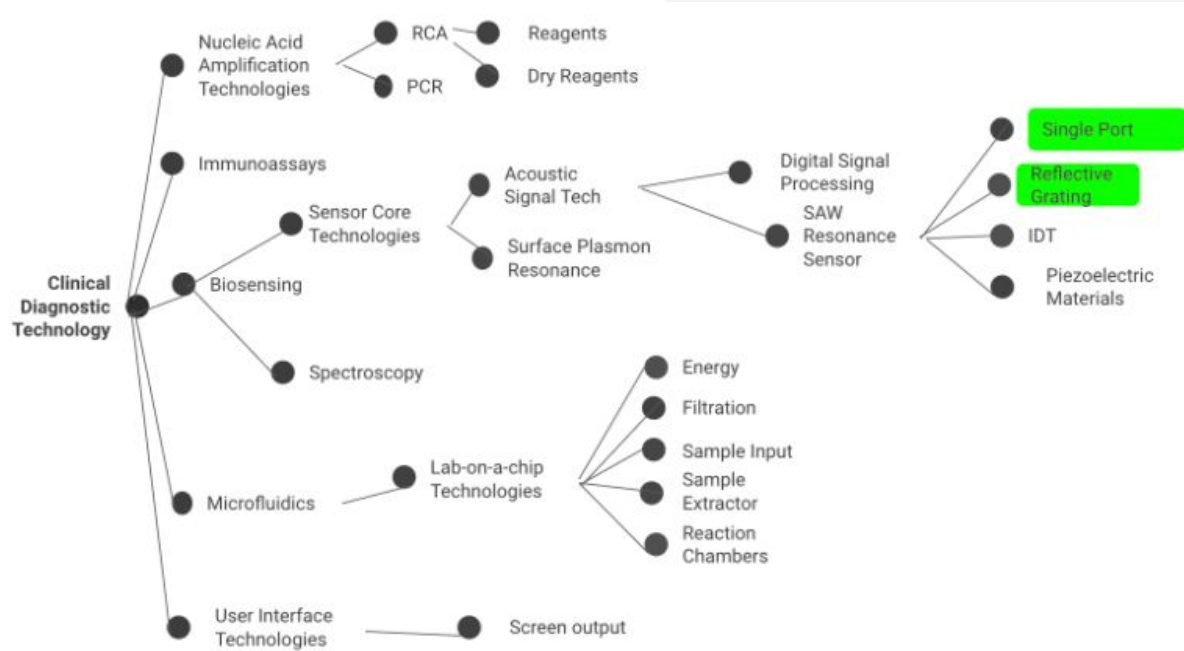
Where does Videm fit in the bigger picture?



SWOT Analysis

Strengths	Weakness	Opportunity	Threats
Handheld/ portable point of care diagnostics that can be completed within 60 minutes	If it is constrained to a regular appointment time, it will require extra personnel or for the doctor to stay longer	The technology can be translated to serve additional markets such as environment, quality control, etc.	Strain workflows by requiring additional staff or longer doctor availability, potentially limiting adoption in healthcare settings where resources are limited
Low cost compared to current laboratory tests	Restructuring of current testing system	Replace time consuming and expensive laboratory tests with hand-held technology	Reliance, trust, and accuracy of current testing methods in the laboratory
The device can provide low cost and high-accuracy diagnosis of up to 10 diseases in one test	Cannot handle high volume of tests/samples at once - would require multiple devices	Provide time sensitive screening and faster time-to-patient care.	Potential emergence of centralized labs or competitors offering higher-throughput diagnostic solutions, which could outcompete the device in settings requiring rapid processing of large patient volumes

Tech Tree



Key Takeaways from our interviews

Caesar Djavaheerian - Carbon Health Chief Innovation Officer and Emergency Physician	-Faster diagnosis will always be desired in medical applications, but will it be just as accurate? -There might be a lot of regulatory hurdles to pass to get this tech to fully replace conventional STI testing -Emergency applications will accept new technology faster: meningitis, MRSA, sepsis, NOT respiratory tract infections
Sara M Sloan - Urgent Care Registered Nurse at Tang Center UC Berkeley	Multiplexing capability and fast diagnosis will be very good in university healthcare systems -there already is Point of care diagnosis but it's more expensive currently -They would be willing to try a few of these devices in parallel to their current testing, but clinics in america are individualized in accepting new technologies so it would be hard to scale without passing FDA approval -Will it handle the volume of STI testing in a university clinic? Will it handle the STI testing of a STI testing clinic? (SF city clinic)

Key Takeaways from our interviews

Sana Elfadil - Clinical Laboratory Manager Tang Center	Many FDA regulations of STI testing will be based on the mechanisms/nature of the test (many of them are specific to antibody tests) so it will be hard to get FDA approval unless something else similar has been done -Given the volume of Tang center testing, this device operating time should be able to handle the amount of tests done.
Dr. Syed Hossainy - Professor, UC Berkeley	Colon cancer and other sorts of cancers w biomarkers in blood and sweat, especially as a fast and cheap way of detection, would be great use cases Specificity to diseases is very important: Having the right biomarkers and a way to detect them
Shruti T, Head of Life Sciences, AvoMD	Partnering up with an already well established venture Can piggyback on their established distribution Can always separate after gaining enough traction
Ezekwe Chinwuba - Engineering Director, Bosch	Biggest challenge is differentiation with the sensor (“It is more so a challenge to get the sensor to detect what you want it to”) - the sensor could detect more than you want or need it to
Cyrill Castro- MEng student, Clinical Operations Specialist LabCorp	- lessening bottlenecks in high-volume labs by handling simpler and urgent tests at the point of care - STI results in 24-48 hours in more centralized labs - allow labs to focus on more complex tests like NGS